

Trainings ▾

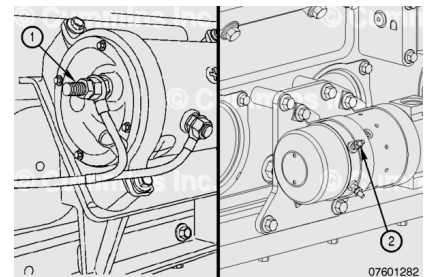
General Information

The engine can be equipped with different pre-lubricating oil systems. This procedure has been written to be generic. The illustrations may **not** match the engine being serviced. Follow the guidelines for the mounting assemblies and hose routings.

There are two different pre-lubricating oil pump designs.

One type is referred to as the "HighVolume" Prelub™ pump. It is distinguished by the ground connection located in the end of the pump body (1).

The other type is referred to as the "Compact" Prelub™ pump. It is identified by having both of the electrical connections located on the side of the pump body (2).



Inspect

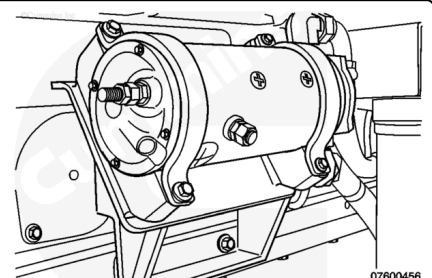
Engines equipped with a pre-lubricating oil system **must not** be operated when the pre-lubricating oil system is **not** working.

Make sure the pre-lubricating oil system wiring is connected.

Refer to Procedure 101-014

(/qs3/pubsys2/xml/en/procedures/28/28-101-014-om.html)

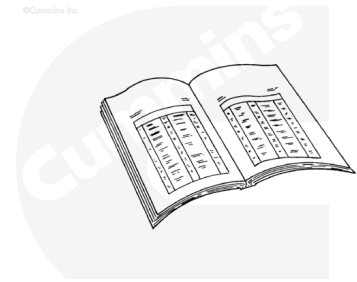
to start the engine. The starting motor will **not** engage until adequate oil pressure is obtained. If the engine has **not** been in operation for some time there will be a delay before the starting motor is engaged. This delay (up to 30-seconds) is normal.



Preparatory Steps



Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Acid is extremely dangerous and can damage the machinery and can also cause serious burns. Always provide a tank of strong soda water as a neutralizing agent when servicing the batteries. Wear goggles and protective clothing to reduce the possibility of serious personal injury.

⚠ WARNING ⚠

On some applications, switches labeled battery disconnect will not disconnect the battery power from the engine starters. Make certain that the battery power has been removed from the engine starters by checking with a multimeter. There must be zero voltage at the starter before beginning this procedure. If there is voltage at the starter, then disconnect the battery terminals directly, removing the negative (-) battery cable first. This must be done to reduce the possibility of personal injury.

⚠ CAUTION ⚠

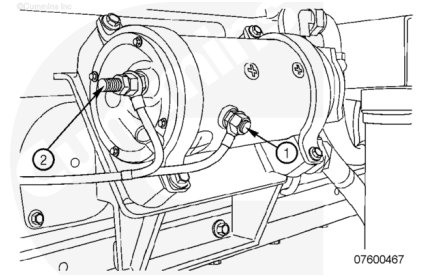
Use caution when draining oil or replacing filters that the oil is not spilled or drained into the bilge area. the oil and oil filters must be disposed of in accordance with local environmental regulations.

- Disconnect the battery cables. Refer to Procedure 013-009 in Section 19 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html>).
- Remove the lubricating oil filters. Refer to Procedure 007-013 in Section 7 (</qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html>).

Remove

The following steps apply to removal of the "High Volume" Prelub™ pump.

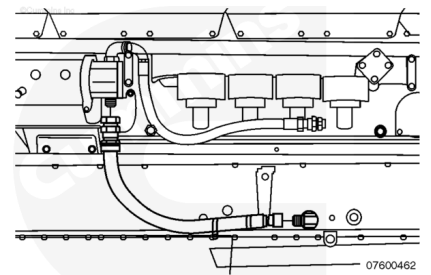
Remove the post timer relay wire (1) and the ground wire (2) from the Prelub™ pump.



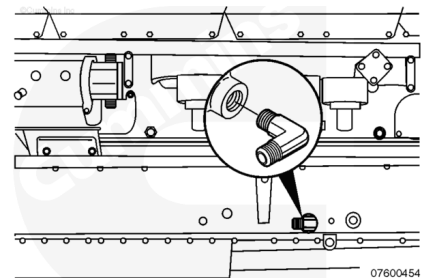
Disconnect the suction hose from the Prelub™ pump.

Disconnect the other end of the suction hose from the elbow located on the oil pan.

Remove the oil pan capscrew, p-clip, and suction hose.

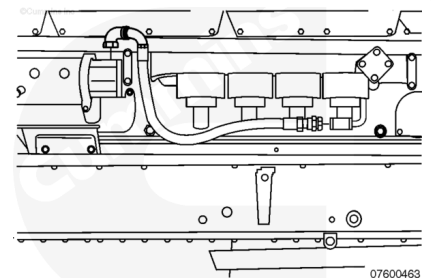


Remove the suction hose elbow.



Disconnect the supply hose from the Prelub™ pump.

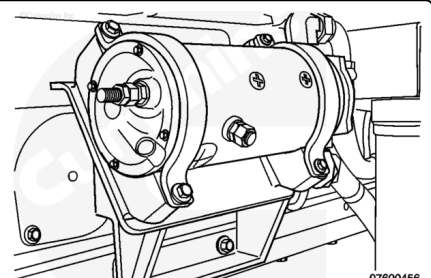
Disconnect the supply hose from the male adapter in the filter head.



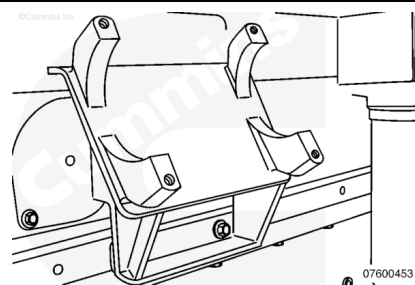
Remove the upper U-shaped pump bracket capscrews.

Loosen the lower U-shaped pump bracket capscrews until the Prelub™ pump can be removed.

Remove the capscrews and U-shaped pump brackets.



Remove the Prelub™ pump bracket mounting capscrews and the bracket.

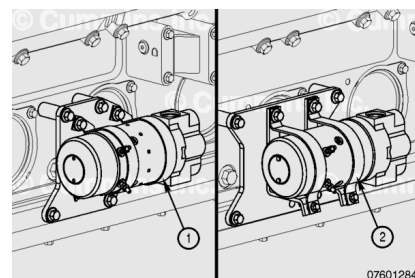


The following steps apply to the "Compact" Prelub™ pump.

The "Compact" Prelub™ pump can be fitted with alternative brackets and clamps, depending on the specific application. The location of the pump on the engine may also vary, but the same procedures can be applied.

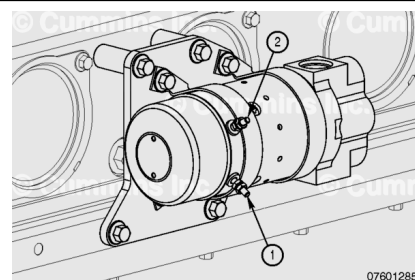
Two of the most common installations are shown in the illustration.

1. Single piece 'loop' clamp
2. Two piece 'U shaped' clamp.



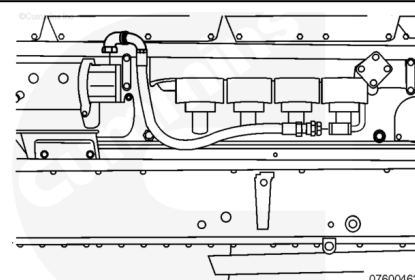
Remove the positive wire (1) and the ground wire (2) from the Prelub™ pump.

Note : The orientation of the pump may vary. The end of the positive connection stud is marked with red paint when new. If this is no longer visible, it is essential to identify which connection is the positive connection, and mark it accordingly to allow for correct assembly.

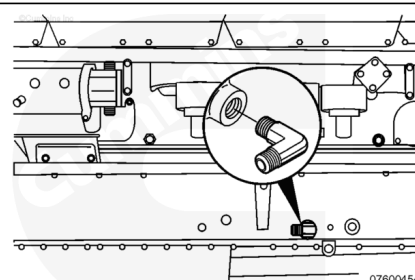


Disconnect the suction hose from the Prelub™ pump.

Remove any hose clips fitted and disconnect the other end of the suction hose from the fitting located in the oil pan.

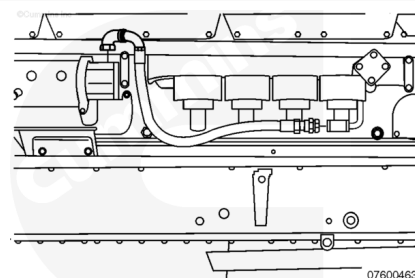


Remove the suction hose elbow.



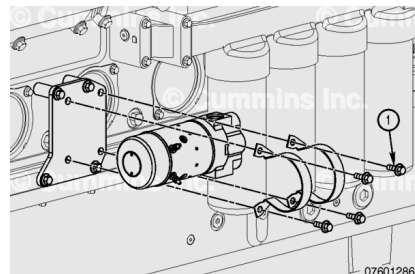
Disconnect the supply hose from the Prelub™ pump.

Remove any hose clips fitted and disconnect the hose from the adaptor in the lubricating oil filter head.



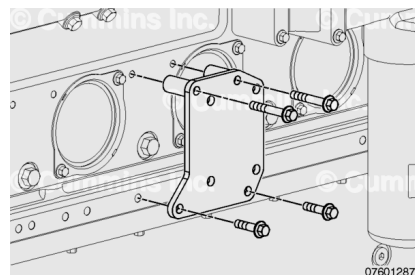
Loosen the clamping capscrews (1) and remove the pump from the engine.

Remove the clamping straps from the mounting bracket.



Remove the capscrews that retain the mounting bracket to the engine block.

Remove the mounting bracket.



Install

The following steps apply to the "High Volume" Prelub™ pump.

Install the bracket and tighten the capscrews.

Torque Value:

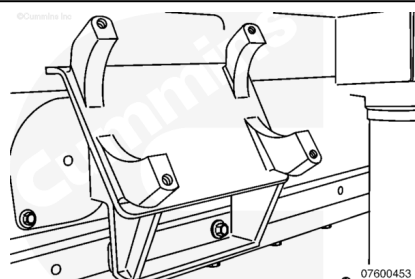
M12 80 n·m [59 ft-lb]

Torque Value:

M10 78 n·m [58 ft-lb]

Torque Value:

3/8 in 45 n·m [33 ft-lb]



Apply Loctite™ 641 to the capscrew threads.

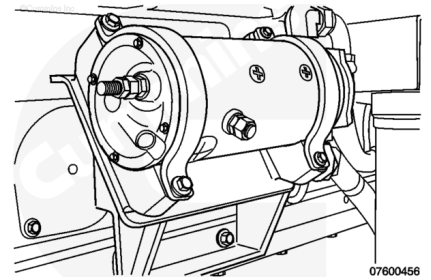
Place the Prelub™ pump into the machined blocks of the bracket.

Place the two U-shaped brackets over the Prelub™ pump.

Install the capscrews and tighten in an alternating sequence.

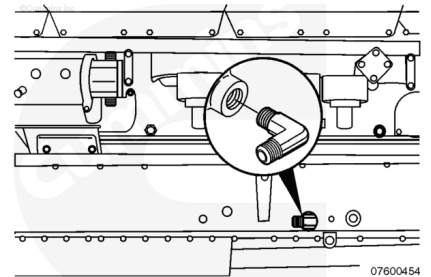
Torque Value: 30 n•m [22 ft-lb]

Make sure the gap is even under each side of the clamp.



Install the elbow into the oil pan adapter and tighten. The final position of the elbow after it has been tightened will be facing toward the rear of the engine.

Torque Value: 74 n•m [55 ft-lb]



Connect the supply hose to the Prelub™ pump.

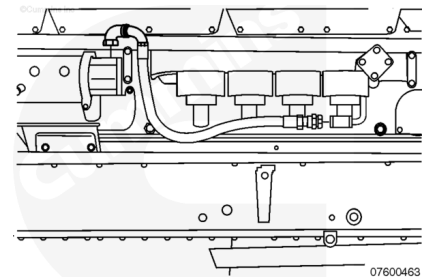
Connect the other end of the hose to the filter head male adapter.

Tighten the fittings.

Torque Value: 163 n•m [120 ft-lb]

Attach the hose to the hose bracket and tighten the capscrew.

Torque Value: 45 n•m [33 ft-lb]



Connect the suction hose to the Prelub™ pump.

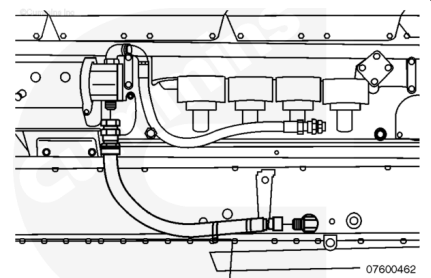
Connect the other end of the suction hose to the elbow in the oil pan rail.

Tighten the fittings.

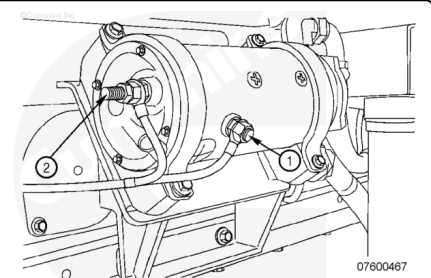
Torque Value: 190 n•m [140 ft-lb]

Attach a p-clip to the hose and secure it to an oil pan adapter capscrew.

Torque Value: 45 n•m [33 ft-lb]



Install the post timer relay wire (1) and the ground wire (2) to the Prelub™ pump.



The following steps apply to the "Compact" Prelub™ pump.

Install the bracket and tighten the capscrews.

Torque Value:

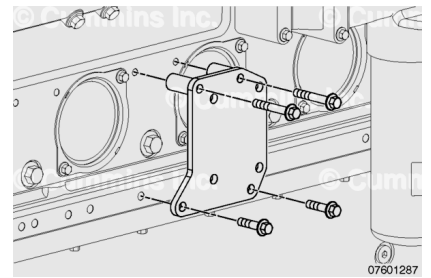
M12 80 n•m [59 ft-lb]

Torque Value:

M11 45 n•m [33 ft-lb]

Torque Value:

3/8 in 45 n•m [33 ft-lb]

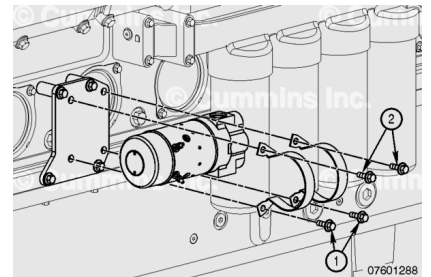


If the single piece 'loop' clamps are installed, loosely fit the lower capscrews (1) retaining the clamps to the mounting bracket.

Install the pump into the clamps. Install the upper capscrews (2).

Tighten the four retaining capscrews in an alternating sequence.

Torque Value: 45 n•m [33 ft-lb]



If the 2 piece 'U shaped' clamp is installed, fit the clamps to the mounting bracket. Tighten **only** the lower capscrews (1).

Torque Value: 45 n•m [33 ft-lb]

Install the pump into the clamps.

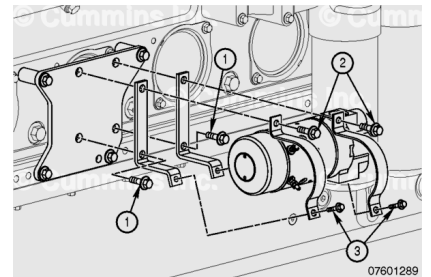
Loosely install the nuts and bolts that join the ends of the clamps (3).

Tighten the upper capscrews (2).

Torque Value: 45 n•m [33 ft-lb]

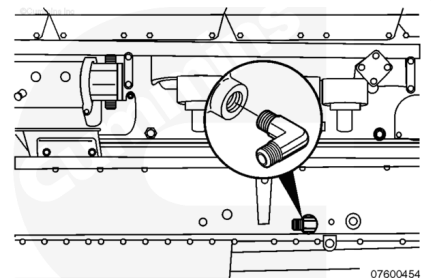
Tighten the clamping nuts and bolts (1).

Torque Value: 30 n•m [22 ft-lb]



Install the elbow into the oil pan and tighten. The final position of the elbow after it has been tightened **must** be correct for the suction tube routing.

Torque Value: 74 n•m [55 ft-lb]

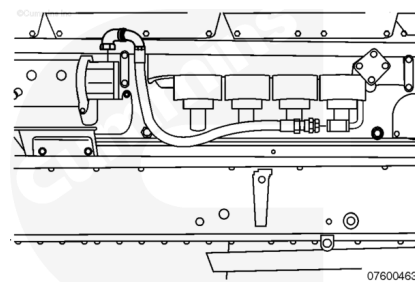


Connect the supply hose to the Prelub™ pump.

Connect the other end of the hose to the filter head male adapter.

Tighten the fittings.

Torque Value: 163 n·m [120 ft-lb]



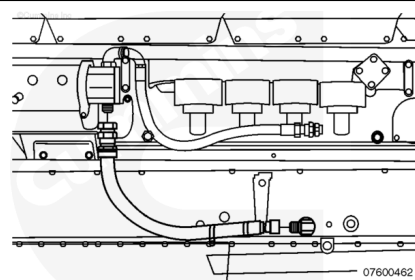
Connect the suction hose to the Prelub™ pump.

Connect the other end of the suction hose to the elbow in the oil pan.

Tighten the fittings.

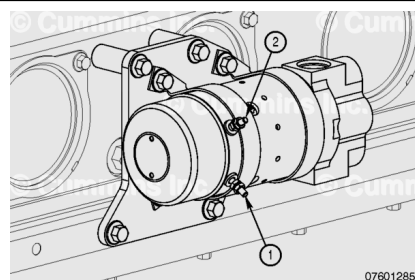
Torque Value: 190 n·m [140 ft-lb]

Replace any hose clips that were removed.



Install the positive wire (1) and the ground wire (2) to the Prelub™ pump.

Note : Make sure there is correct polarity of the connections. The pump may be orientated in the opposite direction compared to the graphic shown.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Acid is extremely dangerous and can damage the machinery and can also cause serious burns. Always provide a tank of strong soda water as a neutralizing agent when servicing the

batteries. Wear goggles and protective clothing to reduce the possibility of serious personal injury.

- Install the lubricating oil filters. Refer to Procedure 007-013 (</qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html>).
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html>).
- Connect the battery cables. Refer to Procedure 013-009 (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>).
- Operate the engine and check for leaks.